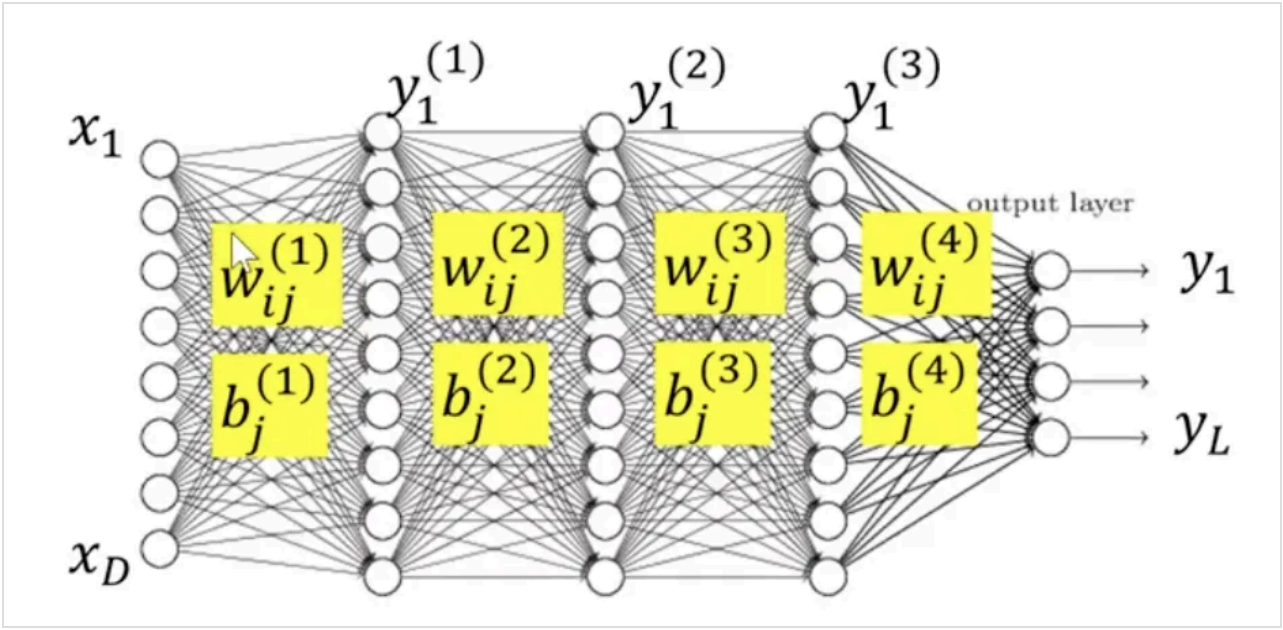
Lec5. Training Part III

How to train network

$$Loss(W) = \frac{1}{T}\sum_{i} div(Y_i, d_i)$$

Minimize $Loss$ w.r.t $\left\{w_{ij}^{(k)}, b_j^{(k)}\right\}$ using gradient descent

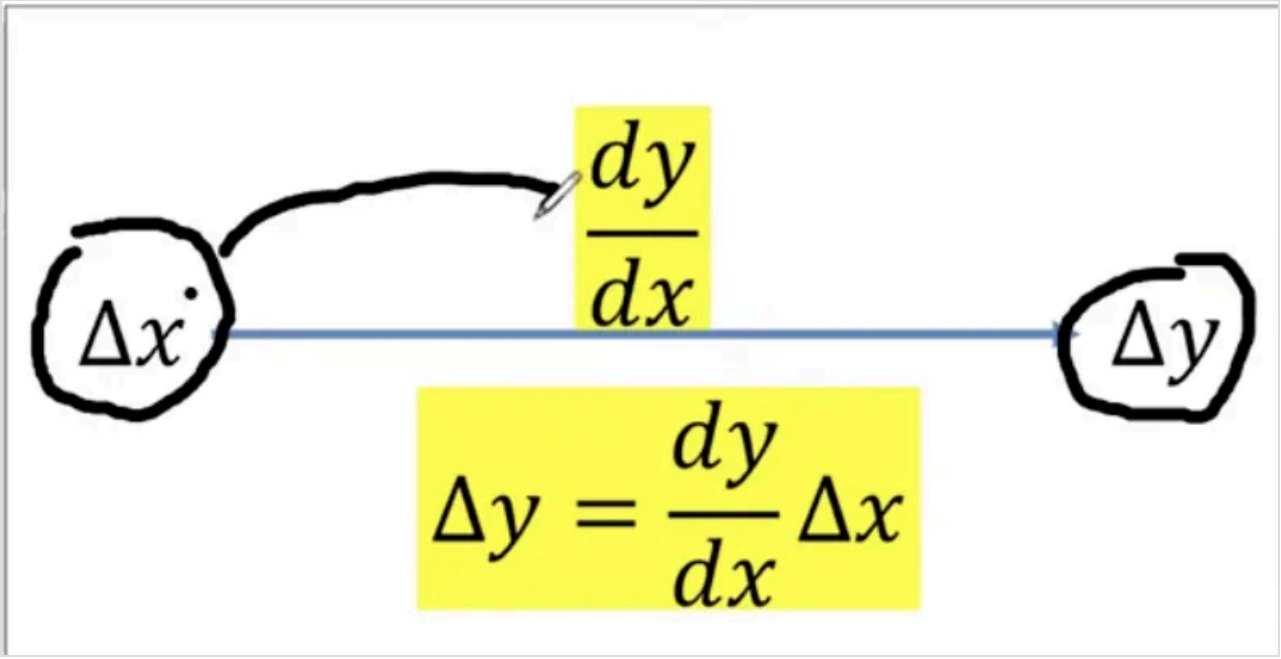
Notation



Influence diagram

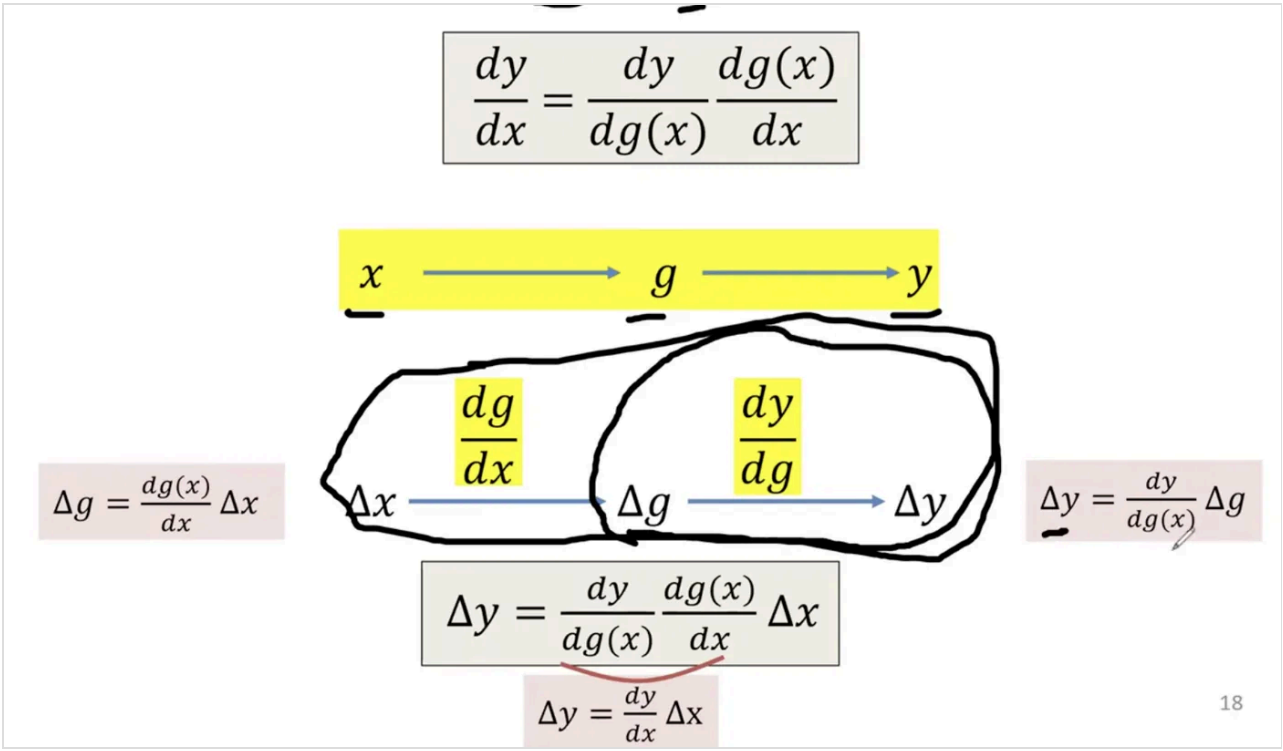


- if x is changed → then y is changed



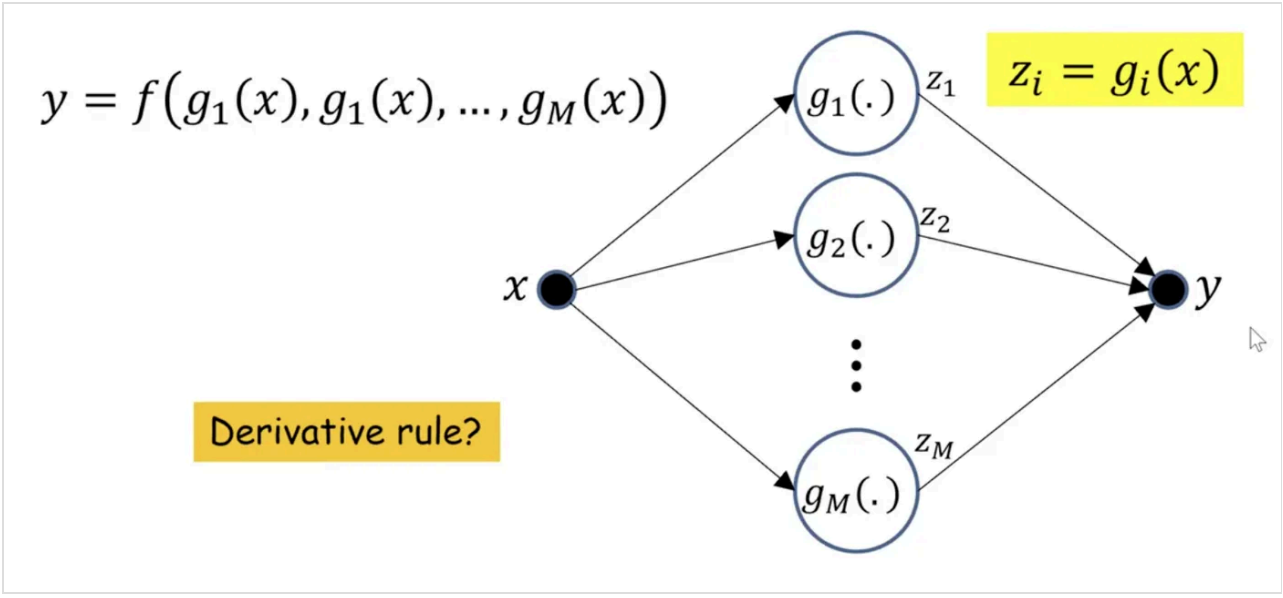
- → simply, $\nabla y = \nabla x * (dy/dx)$

Chain rule



- x에 대한 y의 변화를 보기 위해서는,
  중간에 있는 ∇g를 이용한 **chain rule**을 보면 됨.

Distributed Chain Rule



- 이 경우, x가 변할 때 $g_1, g_2, \ldots$들이 변하므로, 각각에 대한 **편미분의 chain rule**을 구할 수 있어야 함.
  → 아래와 같이

For $y = f(z_1, z_2, \ldots, z_M)$

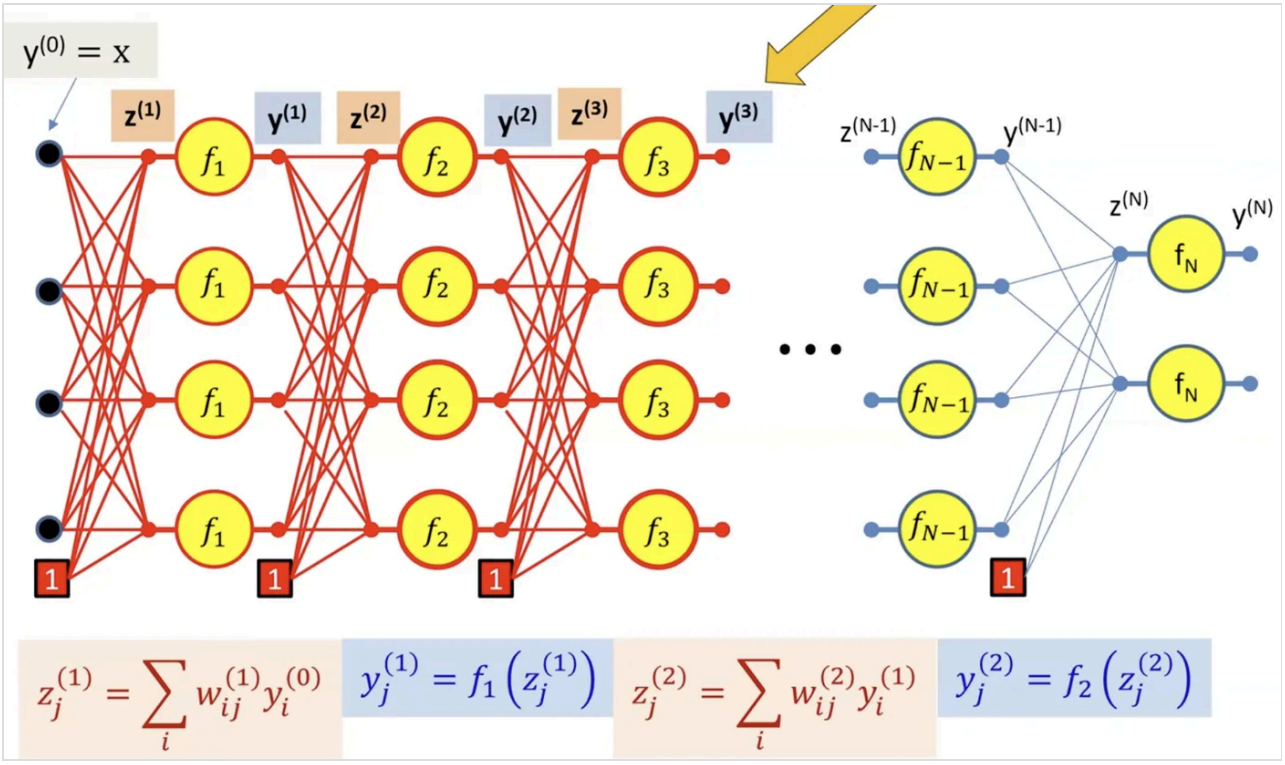
where $z_i = g_i(x)$

$$\Delta y = \sum_i \frac{\partial y}{\partial z_i} \Delta z_i \qquad \Delta z_i = \frac{dz_i}{dx} \Delta x$$

$$\frac{dy}{dx} = \frac{\partial y}{\partial z_1}\frac{dz_1}{dx} + \frac{\partial y}{\partial z_2}\frac{dz_2}{dx} + \cdots + \frac{\partial y}{\partial z_M}\frac{dz_M}{dx}$$

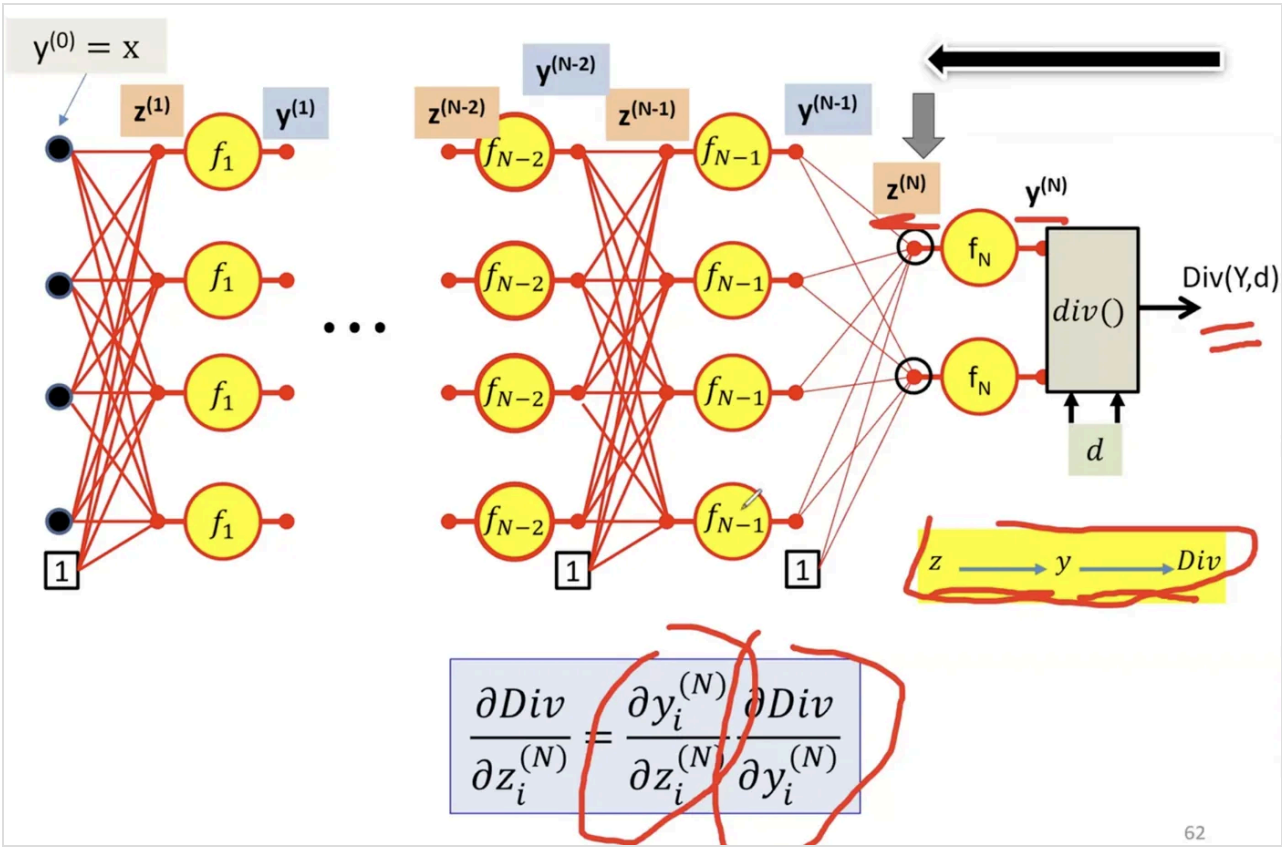
How to train network??

Forward path

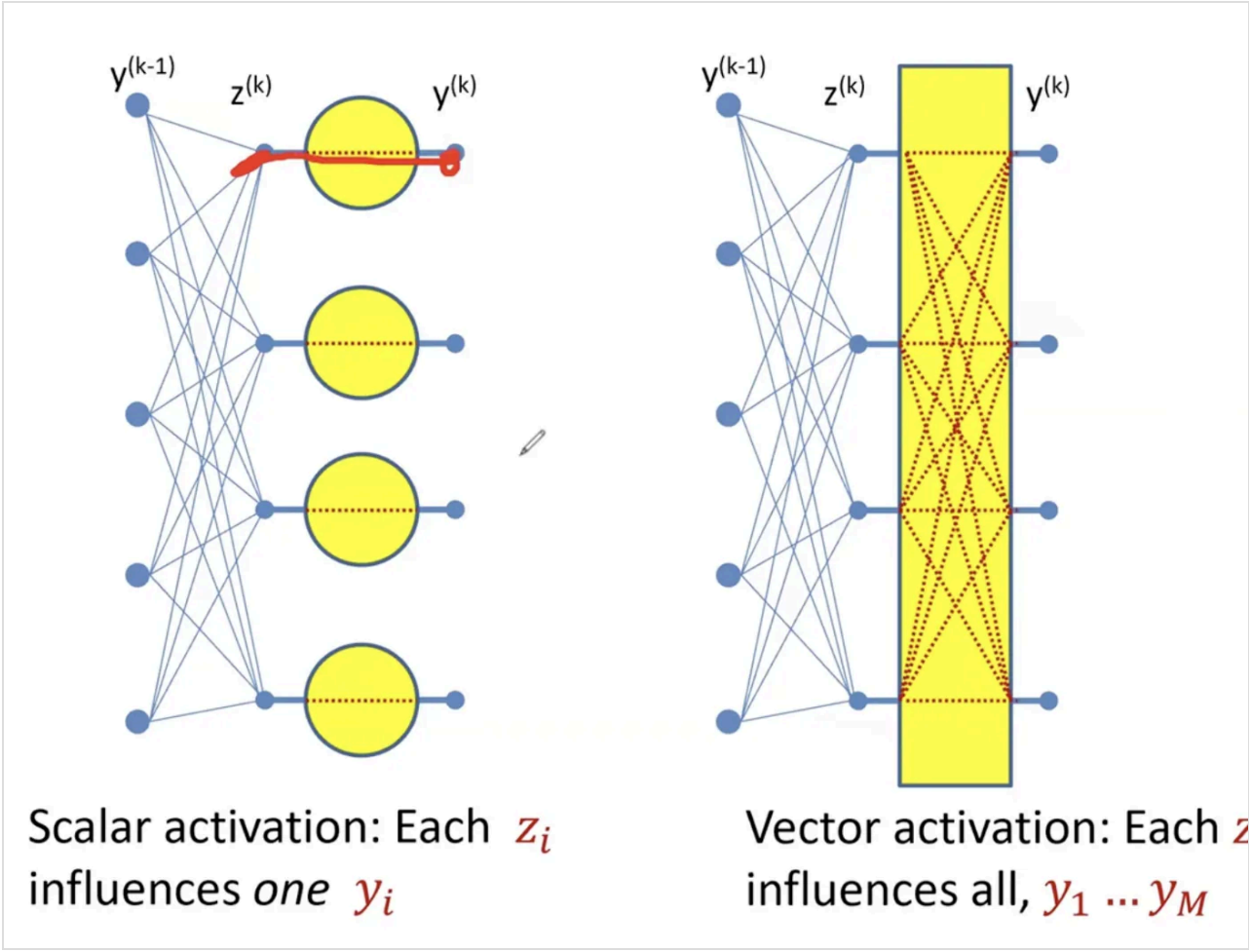


$$z_j^{(1)} = \sum_i w_{ij}^{(1)} y_i^{(0)} \qquad y_j^{(1)} = f_1\left(z_j^{(1)}\right) \qquad z_j^{(2)} = \sum_i w_{ij}^{(2)} y_i^{(1)} \qquad y_j^{(2)} = f_2\left(z_j^{(2)}\right)$$

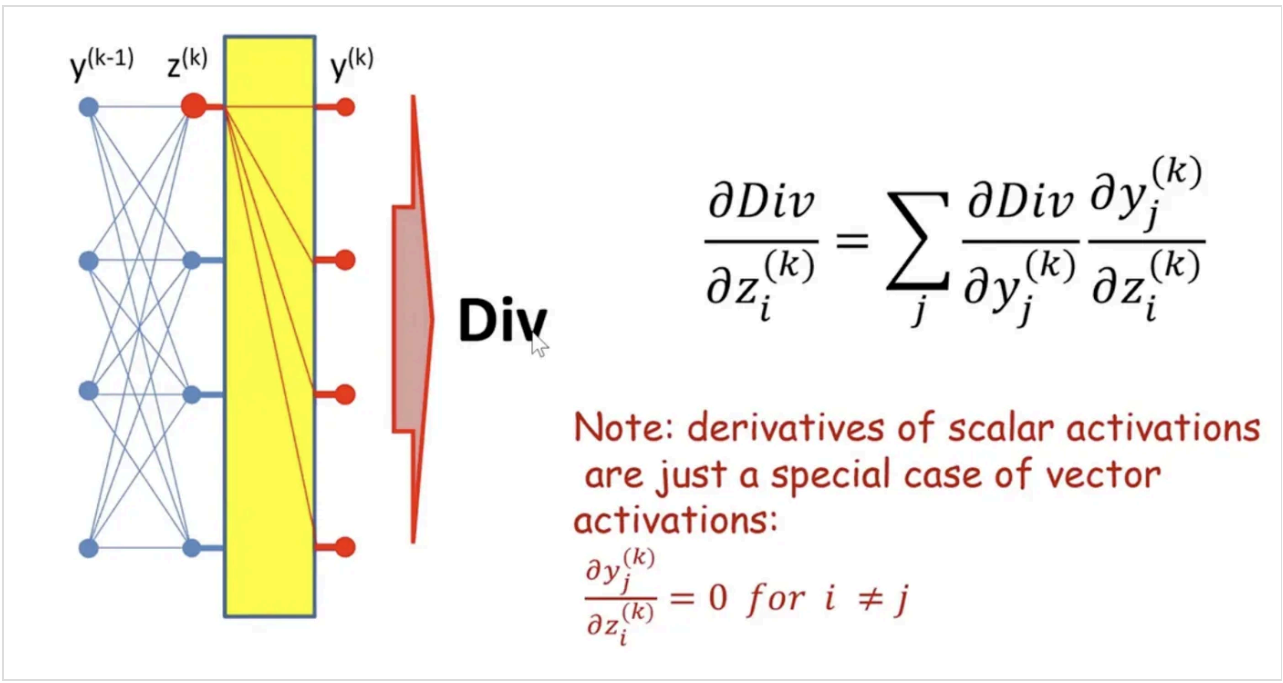
Backward path



Special Case 1. Vector activations



- activation에서, 각 z_i는 모든 y_i에 영향을 주니, 오른쪽과 같은 Influence diagram이 나타나게 됨



- 따라서, derivatives 구할 때, y변화에 대한 영향을 모두 더해야 함.

EX) Softmax

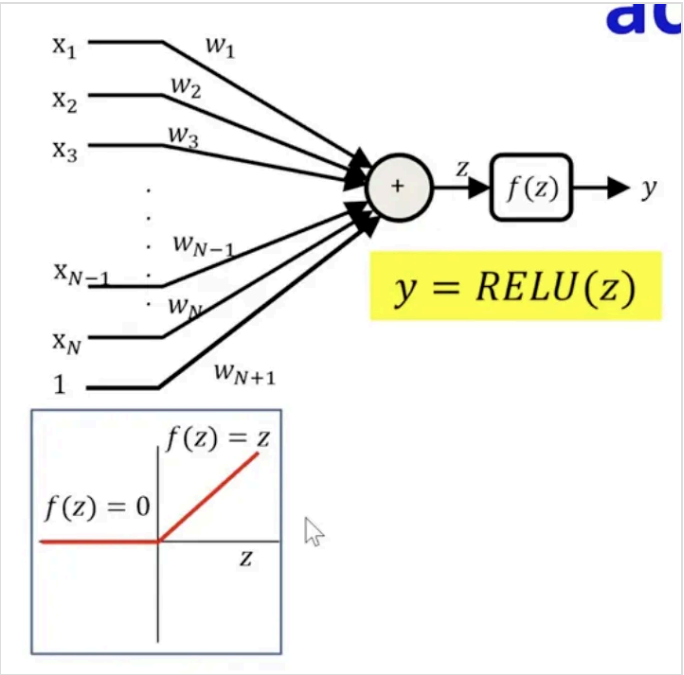
$$y_i^{(k)} = \frac{exp\left(z_i^{(k)}\right)}{\sum_j exp\left(z_j^{(k)}\right)}$$

$$\frac{\partial Div}{\partial z_i^{(k)}} = \sum_j \frac{\partial Div}{\partial y_j^{(k)}} \frac{\partial y_j^{(k)}}{\partial z_i^{(k)}}$$

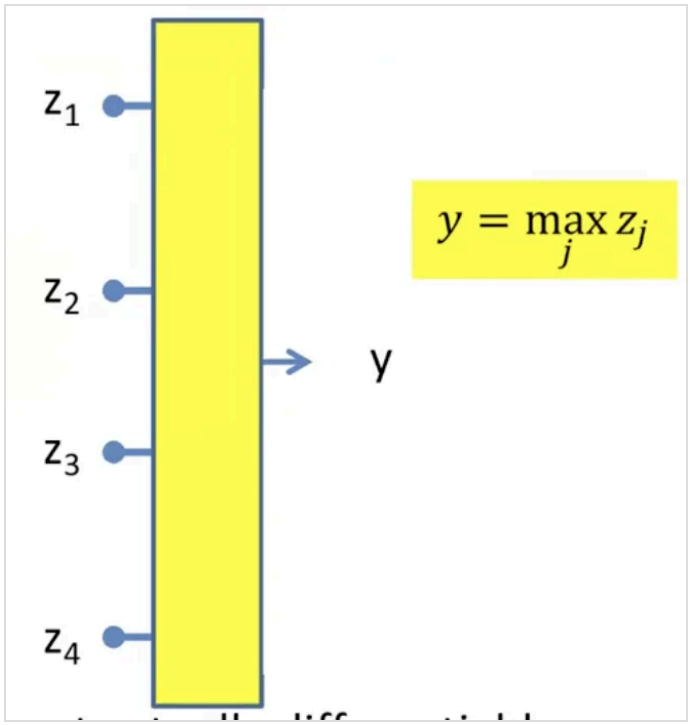
$$\frac{\partial y_j^{(k)}}{\partial z_i^{(k)}} = \begin{cases} y_i^{(k)}\left(1 - y_i^{(k)}\right) & \text{if } i = j \\ -y_i^{(k)} y_j^{(k)} & \text{if } i \neq j \end{cases}$$

Special Case : Non-differentiable activations

ReLU(z)



- 0과 1 사이의 기울기를 사용할 수 있음.
  (ReLU가 그냥 기울기가 매우 빠르게 변하는 함수라고 생각)



$$\frac{\partial y}{\partial z_i} = \begin{cases} 1, & i = \underset{j}{\operatorname{argmax}}\, z_j \\ 0, & otherwise \end{cases}$$

- max의 경우, 가장 큰 값을 주는 j에서만, 미분값 1을 가지게 된다.